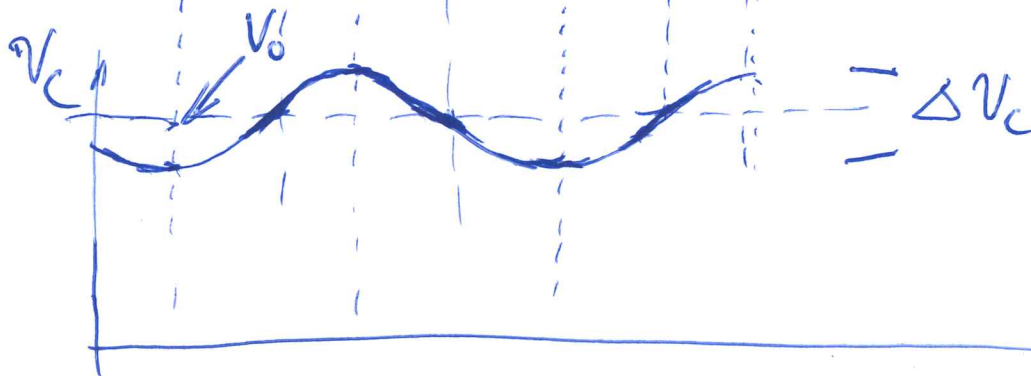
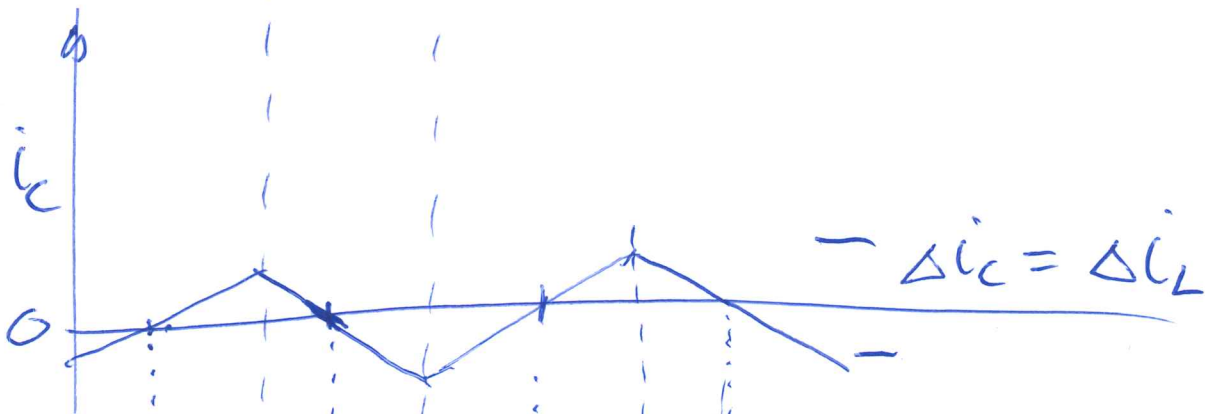
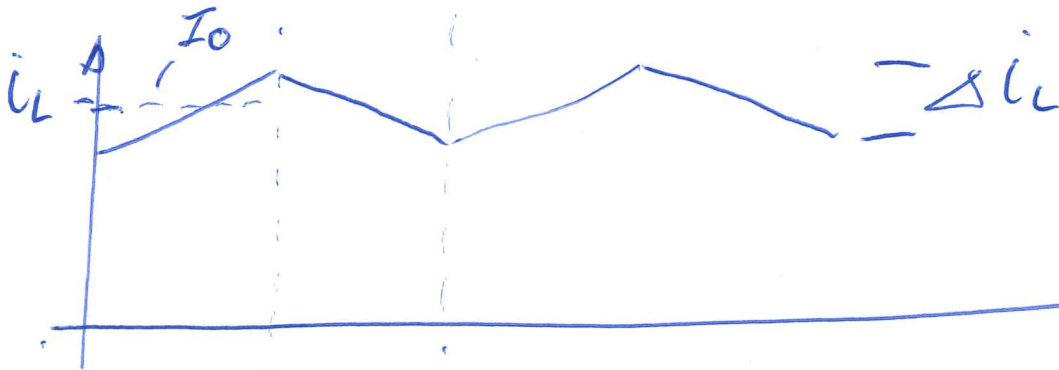
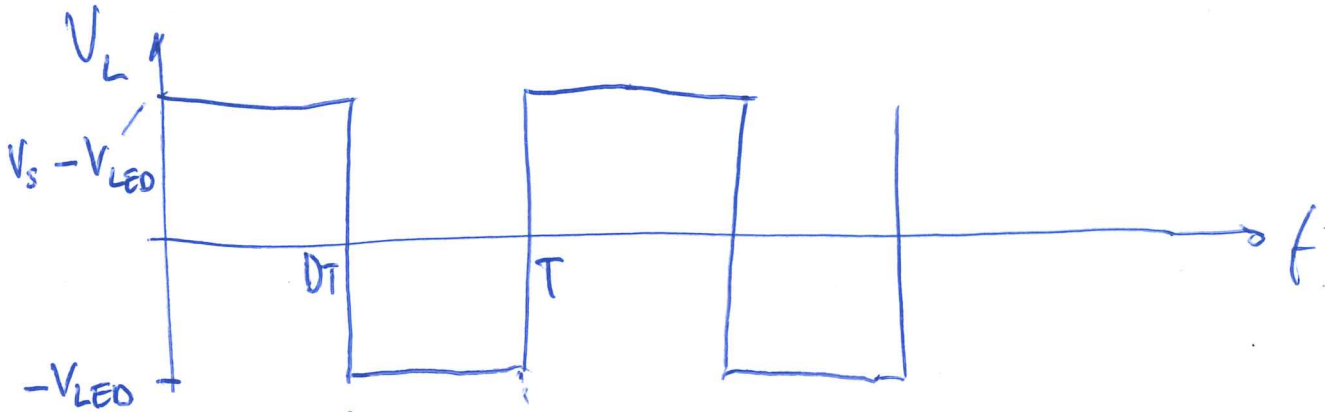
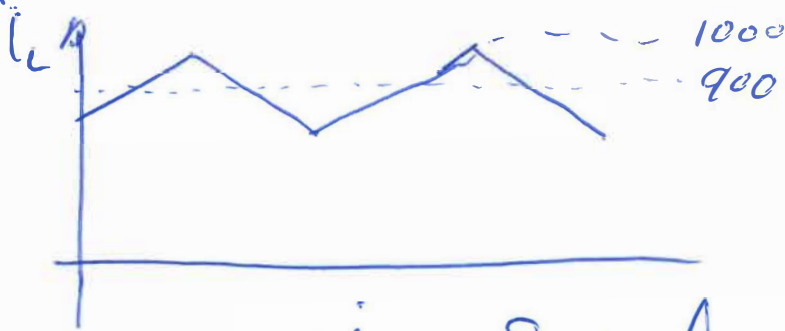


HWk 2 Answers

i) waveforms



(ii) Average LED current is 900 mA (I_0), and peak current is 1000 mA



So $\Delta I_L = 200 \text{ mA}$.

Inductor current ripple $\rightarrow$ for switch off we have

$$\Delta I_L = \frac{V_{LED}(1-D)}{Lf_s}$$

So $L = \frac{V_{LED}(1-D)}{\Delta I_L f_s}$

What is D ?

$$D = \frac{V_o}{V_{in}} = \frac{V_{LED}}{V_s}$$

So either $D = \frac{3.6}{7.2} = 0.5$

or $D = \frac{3.6}{4.8} = 0.75$

Worst case is $D=0.5$ (largest L needed).

So
$$L = \frac{3.6(1-0.5)}{250 \times 10^3 \times 0.2}$$
$$= 36 \mu\text{H}.$$

(iii) Calculate C_2

$$\text{From } \Delta V_s = \frac{I_o D(1-D)}{f_s C}$$

we have

$$C = \frac{I_{LED}(1-D)}{f_s \Delta V_s}$$

Have 2 choices for V_s , but we know the worst case is for $D=0.5$. This D is achieved for $V_s = 7.2 \text{ V}$.

so

$$\begin{aligned} C &= \frac{0.9 \times 0.25}{250 \times 10^3 \times 100 \times 10^{-3}} \\ &= 9 \text{ } \mu\text{F}. \end{aligned}$$